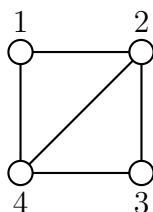


Indian Statistical Institute

BSDS - Second Year

Exercise Set 6.

1. Consider a random walk on the following graph:



Find the probabilities f_{ij}^* for $i, j \in \mathcal{S}$ and also $\mu_i = \mathbb{E}(\tau_i \mid X_0 = i)$.

2. Consider a Markov chain on the state space $\mathcal{S} = \{1, 2, 3, 4\}$ with transition matrix

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} \\ \frac{2}{3} & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{2} & \frac{2}{3} & 0 \end{bmatrix} \end{matrix}.$$

- a) Find the expected return times for each of the states 1, 2, 3, 4.
- b) What is the probability that the chain starting from 1, returns to 1 before visiting the state 4.
- b) What is the probability that the chain starting from 1, visits the state 3 at least twice before returning to state 1.
3. Consider two gamblers A and B, A with initial capital Rs. a and B with initial capital Rs. b . At each time point, first A tosses a coin with success probability p and then B tosses the same coin. If A gets success and B gets failure, then A wins Rs. 1 from B; if A gets failures and B gets success, then B wins Rs. 1 from A. In other cases, no exchange of money happens and the same procedure is repeated. What is the probability that A wins?

4. Consider the Markov Chain on $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$ with the transition matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{2}{5} & \frac{3}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{6} & \frac{1}{3} & 0 & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & 0 & \frac{1}{8} & 0 \end{bmatrix} \end{matrix}.$$

- a) Find the values of f_{33}^* and f_{44}^* .
- b) What is the probability that the chain starting from 3, visits the state 4 before visiting state 6.
- b) Let $\mathcal{C} = \{1, 2\}$. Show that $\mathcal{C}$ is a closed set. Define $\tau_{\mathcal{C}}$ as the first visiting time to $\mathcal{C}$, i.e.,

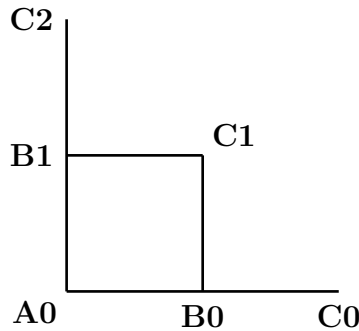
$$\tau_{\mathcal{C}} = \inf\{n \geq 0 : X_n \in \mathcal{C}\}.$$

Define $m_i = \mathbb{E}(\tau_{\mathcal{C}} \mid X_0 = i)$. For $i = 3, 4, 5, 6$, show that m_i satisfies the linear equations

$$m_i = 1 + \sum_{j=3}^6 p_{ij} m_j.$$

Solve the equations.

5. Consider a random walk on the graph



with the modifications that the vertices **C0**, **C1** and **C2** are absorbing states. Find the absorption probabilities at **C0** starting from any vertex.

6. Suppose a fair coin is being flipped until either of the patterns HTH or HHH appear. Find the probability that HTH appears before HHH.

7. You are playing a game which consists of tossing a coin until either one of the patterns of success and failures, one chosen by you and other by your opponent, appear. Assuming that the coin tosses are fair, show that for any pattern of length 3 chosen by your opponent, you can choose a pattern of length 3 so that the game is favourable to you.

[Hint : Representing H by 1 and T by 0, for a pattern $\epsilon_1\epsilon_2\epsilon_3$, chosen by your opponent, choose the pattern $(1 - \epsilon_2)\epsilon_1, \epsilon_2]$

8. Consider the Markov Chain on $\mathcal{S} = \{1, 2, 3, 4\}$ with the transition matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{2}{5} & \frac{3}{5} & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{bmatrix} \end{matrix}.$$

Find all the stationary distributions.

9. Consider the Markov Chain on $\mathcal{S} = \{1, 2, 3, 4\}$ with the transition matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{3}{5} & \frac{2}{5} & \frac{1}{5} & 0 \\ 0 & \frac{1}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{bmatrix} \end{matrix}.$$

Find all the stationary distributions.

10. Consider two independent simple symmetric random walks X_n, Y_n . Consider the joint random walk restricted to $X_n \geq 0, Y_n \geq 0, X_n + Y_n \leq 10$. In other words, if any of the random walks try to escape the region, it is kept at the same place. Can you find the stationary distribution?

11. Consider an irreducible chain having finitely many states. Suppose that for a fixed state $i_0 \in \mathcal{S}$, we have $p_{ii_0} > 0$ for all $i \in \mathcal{S}$. Show that if

$$p_{i_0i}p_{ij}p_{ji_0} = p_{i_0j}p_{ji}p_{ii_0} \text{ for all } i, j \in \mathcal{S},$$

then there is a stationary distribution π and the chain is reversible in π .

[Hint: take $\pi(i) = Cp_{i_0i}/p_{ii_0}$ for a suitable choice of C]